

CSA 1718 – Artificial Intelligence

Name : P. Yaswanth

Reg No : 192524167

Prolog Program to Implement Best First Search Algorithm

Aim

To write and implement a Prolog program to perform a simple Best First Search algorithm.

Objective

- To understand the concept of Best First Search.
- To represent a search graph using Prolog facts.
- To find a path from a starting node to a goal node.

Theory

Best First Search is a search technique that selects a promising node based on a cost or evaluation value.

In this example, the edges are given with costs. The algorithm selects the available node with the smaller cost first.

Algorithm

1. Define the graph using edge(Node1, Node2, Cost).
2. Specify the starting node and goal node.
3. Find the neighboring nodes.
4. Sort the nodes according to their cost.
5. Select the lowest-cost node.
6. Continue the search until the goal is reached.
7. Display the path.

Program

edge(a, b, 4).

edge(a, c, 2).

edge(b, d, 5).

edge(c, d, 1).

edge(d, e, 3).

best_first(Start, Goal, Path) :-

search([Start], Goal, Path).

search([Goal|_], Goal, [Goal]).

search([Current|Rest], Goal, [Current|Path]) :-

findall(Cost-Node,

(edge(Current, Node, Cost),

\+ member(Node, [Current|Rest])),

Children),

sort(Children, Sorted),

nodes(Sorted, NewNodes),

append(NewNodes, Rest, NewQueue),

search(NewQueue, Goal, Path).

nodes([], []).

nodes([_Node|T], [Node|Rest]) :-

nodes(T, Rest).

Query

best_first(a, e, Path).

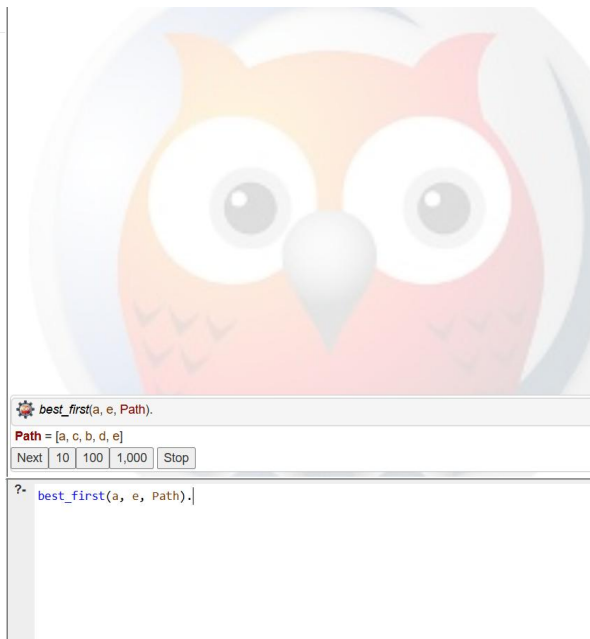
Output

Path = [a, c, d, e].

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1 edge(a, b, 4).
2 edge(a, c, 2).
3 edge(b, d, 5).
4 edge(c, d, 1).
5 edge(d, e, 3).
6
7 best_first(Start, Goal, Path) :-
8     search([Start], Goal, Path).
9
10 search([Goal|_], Goal, [Goal]).
11
12 search([Current|Rest], Goal, Path) :-
13     findall(cost-Node,
14         (edge(Current, Node, Cost),
15          \+ member(Node, [Current|Rest])),
16         Children),
17     sort(Children, Sorted),
18     add_nodes(Sorted, Rest, NewQueue),
19     search(NewQueue, Goal, RestPath),
20     Path = [Current|RestPath].
21
22 add_nodes([], Queue, Queue).
23
24 add_nodes([_Node|T], Queue, NewQueue) :-
25     append(Queue, [Node], Temp),
26     add_nodes(T, Temp, NewQueue).

```



Result

Thus, the Best First Search algorithm was successfully implemented in Prolog and a path from the starting node to the goal node was obtained.